End Semester Examination 2025

Name of the Program: B. Tech. (EE) | Semester: IVth

Name of the Course: Mathematics III | Course Code: **TMA 401**

Time: 3 Hours | Maximum Marks: 100

Note:

(i) All Questions are compulsory

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks in each main question are twenty

(iv) Each question carries 10 marks

Q1	**(20 marks)**	
a)	In a competitive examination, 30 candidates are to be selected. In all 600 candidates appear in the written test, and 100 will be called for interview. What is the probability that a person will be called for an interview? Find the probability of a person getting selected, if he has called for an interview.	
b)	It is known that 4% of the batch of components in a manufacturing process are faulty components are tested on production line with 90% probability that a faulty components detected, but it is known that in 2% of cases a component which is not faulty is nevertheless recorded as faulty. What is the probability that a component which is recorded as faulty is actually faulty?	**CO 1**
c)	There are two bags. In the first bag there are three white and two black balls and in the second, one white and 4 black balls. One of the bags is selected and one ball is drawn. It is shown to be white. what is the probability that it came from the second bag.	
Q2	**(20 marks)**	
a)	The diameter of an water pipe, say X , is assumed to be a continuous random variable with probability density function $f(x) = 6x(1-x), 0 \le x \le 1$. **(i)** Check validity of f(x) to be *p.d.f.* ***(ii)*** Determine a number b such that $P(X < b) = P(X > b)$.	
b)	If X is uniformly distribution over (0, 10), Find (i) $P[X < 30]$, (ii) $P[3 < X < 8]$	**CO 2**
c)	The mean and variance of a binomial variate X with parameters n and p are 16 and 8. Find the following: (i) $P(X = 0)$ (ii) $P(X = 1)$ (iii) $P(X \ge 2)$.	
Q3	**(20 marks)**	
a)	The joint density of X and Y is $f(x,y) = \frac{x+2y}{27}$ where x and y can be assumed only integer value 0, 1, and 2. Find the conditional distribution of Y for X=x.	
b)	If $f(x,y) = \begin{cases} e^{-(x+y)}, & x \ge 0, y \ge 0 \\ 0, & elsewhere \end{cases}$ is joint probability density function of random variable X and Y, Find (i) P[X<1] (ii) P[X+Y<1]	**CO 3**

c) If X and Y have join p.d.f

Y \ X	1	3	9
2	1/8	1/24	1/12
4	¼	¼	0
6	1/8	1/24	1/12

Find

(i) Marginal distribution of X and Y

(ii) Conditional distribution of X given Y=2

(iii) Find covariance of X and Y?

Q4 **(20 marks)**

a) Toss 3 coins. Let X denotes the number of heads on the first two and Y denotes the Number of heads on the last two.

(i) Find the joint distribution of X and Y

(ii) Find $E[y / x = 1]$

(iii) Find the correlation coefficient $\rho(x, y)$

b) In Miss India Contest Ten competitor are ranked by three judges in the given table. Use the correlation coefficient to determine which pair judge has the nearest approach to common perception in beauty. **CO 4**

1st Judge	5	3	10	7	2	1	4	10	4	6
2nd Judge	1	6	5	10	3	2	4	9	7	8
3rd Judge	6	4	9	8	1	2	3	10	5	7

c) Find the coefficient of skewness and kurtosis from the following and conclude your result:

x	0	1	2	3	4	5	6	7	8
f	1	8	28	56	70	56	28	8	1

Q5 **(20 marks)**

a) A random sample of 16 values from a normal population showed a mean of 41.5 inches and a sum of square of deviation from this mean equal to 135 inches. Show that the assumption of a population mean of 43.5 inches is not valid at 5% level of significance.(for two tailed test $t_{15}(0.025) = 2.13$).

b) The following data relate to the number of limit producers per shift by two workers A and B a number of day: **CO 5**

A	19	22	24	27	24	18	20	19	25		
B	26	37	40	35	30	30	40	26	30	35	45

Can it be inferred that A is more stable worker compared to B?

Answer using F-test at 5% level of significance. *{From Table $F_{10,8}(0.05) = 3.35$}*

c) In an experiment on pea breeding. Mendal obtained the following frequencies of seeds, 315 round and yellow; 101 wrinkled and yellow; 108 round and green; 32 wrinkled and green; total 556. Theory predicts that the frequence would be in the proportion 9:3:3:1. Does the experimental results support the theory? *{From Table* $\chi_3^2(0.05) = 7.185$*}*